

ATTACK ON TITAN COMPENDIUM

COMPLETE TECHNICAL ARCHITECTURE, LORE CODEX & FEATURE DOSSIER

Project:	Attack on Titan Interactive Lore Compendium & Tactical Archive
Author / Chief Archivist:	ADITYA (Chief System Architect)
Repository:	https://github.com/barnwaladitya66-eng/Attack-On-Titan.git
Target Milestone:	Commit b604d57 — Razor-Sharp High-Impact Scroll Triggers & OLED Blood Black
Core Theme:	Permanent Blood Black OLED (#000000, #ff0033, Smoked Ruby, 100% Blur-Free)
Document Scope:	System Architecture, 2,000-Year Timeline, 9 Titans Codex, 12 Character Dossiers, World & Walls Radar, Battle Matrix, Lineages, and Tactical Systems

EXECUTIVE SUMMARY

The **Attack on Titan Compendium** is a comprehensive, production-grade lore archive and tactical intelligence system created for fans and scholars of Hajime Isayama's masterwork. Built with modern web standards (pure semantic HTML5, high-performance CSS3 animations, and modular vanilla ES6 JavaScript), the project presents 2,000 years of Eldian history, the Nine Titan Shifters, concentric territorial wall specifications, and tactical military dossiers in an immersive, high-contrast dark aesthetic.

This official documentation records every technical feature, lore data structure, visual design milestone, and interactive controller engineered from project inception up through **Commit b604d57** ('Remove blur effect from scroll triggers keeping ultra-crisp 80px glide and scale pop').

1. SYSTEM ARCHITECTURE & VISUAL DESIGN SYSTEM

1.1 Multi-Page Architecture & URL Routing

The application is structured into five distinct interconnected portals, sharing a unified CSS variable design system, dynamic particle canvas, global modal controller, and scroll trigger systems:

Portal File	Title & Functionality	Key Interactive Systems
index.html	Command Portal & Strategic Overview	Hero banner, Concentric Walls Radar, Featured Titans, Scout dossiers, stats counter
timeline.html	2,000-Year Historical Paths Timeline	Chronological timeline feed, Paths Ancestral Tree visualizer, Rumbling Simulator
titans.html	The Nine Titan Shifters Codex	9 Titans catalog, Battle Matchup Simulator, Lineage flowchart, Height Comparator
characters.html	Military Personnel Intelligence Dossiers	12 Character cards, Combat radar meters, Japanese voice lines, Allegiance filter
world.html	Territory, Walls & Military Equipment	Wall Maria/Rose/Sina specs, ODM gear mechanics, Tactical Satellite Map, Public Disclosures

1.2 The Permanent 'Blood Black' OLED Theme System

Following iterative UI experimentation, the visual aesthetic was locked to the permanent **Blood Black OLED Theme**, configured via CSS custom properties in `css/style.css`:

CSS Token	Value / Hex	Design Application
--bg-primary	#000000	Absolute True OLED pitch-black backdrop for maximum contrast.
--bg-card	rgba(16, 2, 5, 0.82)	Smoked ruby translucent glass cards with 1px border rgba(255, 0, 51, 0.25).
--accent-red	#ff0033 / #e63946	Neon crimson glow for badges, active states, buttons, and combat highlights.
--accent-gold	#d4af37 / #ffd700	Regal metallic gold for royal emblems, Japanese kanji quotes, and clearances.
--font-display	Cinzel, serif	Classical epic typography for main headers and anime arc titles.
--font-tech	Share Tech Mono, mono	Military telemetry, coordinates, threat ratings, and technical specs.

1.3 High-Impact Razor-Sharp Scroll Trigger Animation Engine (Commit b604d57)

In **Commit b604d57**, all blur filters were removed to ensure 100% razor-sharp clarity. The system uses an IntersectionObserver with high-inertia spring easing:

- **80px Entrance Glide:** Pre-scroll elements sit 80px below at 0.92 scale (opacity: 0; transform: translateY(80px) scale(0.92)) and smoothly glide up to full size (opacity: 1; transform: translateY(0) scale(1)) over 0.95s cubic-bezier(0.16, 1, 0.3, 1).
- **Directional Variants:** .reveal-left (-100px slide), .reveal-right (+100px slide), .reveal-scale (zoom from 0.78).
- **Cascading Wave Delays:** .stagger-1 (0.15s) through .stagger-6 (0.90s) for fluid multi-column entrance animations.
- **Animated Stat Counters:** Numbers (2,000+, 9, 3, 80%) smoothly count up from 0 when crossing into view.
- **Floating Back-to-Top Button:** Automatically appears after scrolling 350px.

2. 2,000-YEAR HISTORICAL TIMELINE & PATHS NEXUS

The historical timeline dataset in js/data.js captures the complete 2,000-year canonical story arc with Japanese quotes and historical impact analysis:

Era & Year	Title & Summary	Canonical Significance & Quote
Origin Era 2,000 Years Ago	Ancient History & Genesis of Titans Slave girl Ymir Fritz bonds with the organic anomaly, gaining Titan powers. King Fritz exploits her to subjugate Marley and establish Eldian empire.	Impact: Birth of Nine Titans, Paths dimension, and Eldian dominance. <i>'To you, 2,000 years from now...'</i>
Cataclysm c. 743 (100 Yrs Ago)	The Great Titan War & Wall Creation King Karl Fritz conspires with the Tyburs to orchestrate Eldia's fall, retreating to Paradis with millions of Colossal Titans inside Maria, Rose, and Sina.	Impact: Creation of the Three Walls, memory wipe, and pacifist vow. <i>'If Eldia is to perish, I will accept it.'</i>
The Breach Year 845	The Fall of Shiganshina Colossal (Bertholdt) and Armored (Reiner) Titans breach Wall Maria. Carla Yeager devoured. Grisha takes Founding Titan and passes to Eren.	Impact: Loss of Wall Maria, 250k casualties, and Eren's sacred extermination vow. <i>'On that day, mankind received a grim reminder.'</i>
First Victory Year 850	Struggle for Trost & Female Titan Eren unlocks Attack Titan, seals Trost gate with boulder. Female Titan (Annie Leonhart) attacks 57th Expedition and is captured in Stohess.	Impact: Discovery of Titan shifters in military ranks; Special Operations Squad formed. <i>'If you don't fight, you can't win. Fight!'</i>
The Truth Year 850	Uprising & Return to Shiganshina Historia crowned true Queen. Scouts retake Wall Maria; Erwin leads suicide charge; Armin gains Colossal Titan; Grisha's basement reveals human civilization thrives outside.	Impact: Wall Maria liberated, Survey Corps reduced to 9, truth of outside world revealed. <i>'My soldiers, rage! My soldiers, scream!'</i>
Global War Year 854	Raid on Liberio & War for Paradis Eren infiltrates Marley, devastates Willy Tybur's festival, devours War Hammer Titan. Yeagerist coup paralyzes military command.	Impact: War Hammer acquired, naval fleet destroyed, Marley counter-invades Paradis. <i>'I will keep moving forward... until enemies destroyed.'</i>
The Finale Year 854	The Rumbling & Battle of Heaven/Earth Eren frees Founder Ymir and unhardens millions of Wall Titans, wiping 80% of humanity. Mikasa decapitates Eren; Titan powers permanently vanish.	Impact: 80% global humanity wiped, Titan Curse eliminated forever. <i>'To you, in 2,000... or 20,000 years.'</i>

2.2 Interactive Paths Ancestral Tree & Rumbling Simulator

- **The Paths Tree:** Metaphysical Coordinate tree connecting Founder Ymir Fritz trunk with branches for the Royal Lineage, Attack Titan, Nine Shifters, and Subjects of Ymir.
- **The Rumbling Simulator:** Phase-by-phase tactical telemetry detailing the unhardening of Wall Maria/Rose/Sina, the 50 km/h march, 600°C steam shockwaves, and the coastal destruction of Marleyan defenses.

3. THE NINE TITAN SHIFTERS CODEX & BATTLE MATRIX

3.1 Complete Specifications of the Nine Titans

Every Titan shifter entity is documented with height, threat ratings, inheritors, and specialized biological traits:

Titan Name	Height & Threat	Known Inheritors	Combat Doctrine & Unique Powers
Founding Titan Shiso no Kyojin	Up to 500m+ Extinction Level	Ymir Fritz, Karl Fritz, Uri Reiss, Frieda, Grisha, Eren	The Coordinate: Absolute command over Titans/Eldians; memory alteration; biological reconstruction.
Attack Titan Shingeki no Kyojin	15 meters Ultra-High	Eren Kruger, Grisha Yeager, Eren Yeager	Future Memory Perception: Accesses past & future inheritor memories; berserk drive; localized hardening.
Colossal Titan Choogata Kyojin	60 meters Catastrophic	Bertholdt Hoover, Armin Arlert	Nuclear transformation detonation; 600°C scalding steam screen emission; massive physical siege destruction.
Armored Titan Yoroi no Kyojin	15 meters Very High	Reiner Braun	Organic hardened armor plating; battering ram kinetic charge; armor shedding for mobility bursts.
Female Titan Megata no Kyojin	14 meters Very High	Annie Leonhart	Titan attraction scream; diamond crystal hardening; high agility and Muay Thai martial arts adaptability.
Beast Titan Kemono no Kyojin	17 meters Catastrophic	Tom Ksaver (Sheep), Zeke Yeager (Ape)	Supersonic boulder artillery pitching; vocal Titan creation & command (Royal Blood); strategic command.
Jaw Titan Agito no Kyojin	5 meters High	Marcel, Ymir, Porco Galliard, Falco Grice	Diamond-dense crushing jaws & claws; extreme omnidirectional speed; feathered flight wings (Falco).
Cart Titan Shariki no Kyojin	4 meters Tactical Support	Pieck Finger	Extreme operational endurance (months shifted); heavy machine gun turret / panzer unit modular harness.
War Hammer Titan Sentsui no Kyojin	15 meters Ultra-High	Lara Tybur, Eren Yeager	Structural crystal weapon generation (warhammers, pikes, crossbows); remote pilot cable capsule.

3.2 Interactive Battle Matrix Duels

The Battle Matrix simulates canonical duels with odds calculations and strategic factors:

- **Captain Levi vs. Beast Titan (Shiganshina):** 94% Levi advantage via supersonic ODM flank and blade blinding.
- **Attack Titan vs. Armored Titan (Trost/Shiganshina):** 65% Eren advantage via Brazilian Jiu-Jitsu grappling and Thunder Spears.
- **Attack Titan vs. War Hammer Titan (Liberio):** 52% Eren advantage utilizing the Jaw Titan's claws as a nutcracker.
- **Eren & Mikasa vs. Colossal Titan (Shiganshina):** 78% Scouts advantage via steam diversion bait and nape hook extraction.
- **Allied Coalition vs. Doomsday Founding Titan (Fort Salta):** 61% Alliance advantage via Falco's flight and Zeke's sacrifice.

4. PERSONNEL MILITARY INTELLIGENCE DOSSIERS

The compendium features 12 fully articulated personnel dossiers with 4-dimensional combat radar stats (Combat, Intelligence, Agility, Willpower) and authentic Japanese quotes:

Personnel Name	Role & Allegiance	Combat Stats (C/I/A/W)	Canonical Quote & Meaning
Eren Yeager Attack & Founder	Attack, Founding & War Hammer Titan Survey Corps / Yeagerists	Combat: 98% Intel: 88% Agility: 90% Willpower: 100%	<i>'Because I was born into this world!'</i> (Ore ga kono yo ni umareta kara da!)
Mikasa Ackerman Humanity's Vanguard	Elite Soldier / Ackerman Vanguard Survey Corps Special Squad	Combat: 100% Intel: 85% Agility: 99% Willpower: 96%	<i>'This world is cruel... and yet so beautiful.'</i> (Kono sekai wa zankoku da... soshite utsukushii)
Armin Arlert 15th Commander	15th Commander / Colossal Titan Survey Corps High Command	Combat: 82% Intel: 100% Agility: 75% Willpower: 92%	<i>'Someone who cannot sacrifice anything can't change anything.'</i>
Captain Levi Humanity's Strongest	Humanity's Strongest Soldier Special Operations Squad Leader	Combat: 100% Intel: 90% Agility: 100% Willpower: 95%	<i>'Choose for yourself whichever decision leaves no regrets.'</i>
Erwin Smith 13th Commander	13th Commander of Survey Corps Survey Corps Supreme Commander	Combat: 90% Intel: 100% Agility: 85% Willpower: 100%	<i>'My soldiers, rage! My soldiers, scream! My soldiers, fight!'</i>
Hange Zoe 14th Commander	14th Commander / Chief Scientist Survey Corps Research Division	Combat: 88% Intel: 98% Agility: 88% Willpower: 94%	<i>'Titans really are magnificent!'</i>
Reiner Braun Armored Titan	Armored Titan / Vice-Captain Marley Warrior Unit	Combat: 92% Intel: 78% Agility: 80% Willpower: 86%	<i>'The real battle starts now... Eren!'</i>
Zeke Yeager Beast Titan	Beast Titan / War Chief Marley Warriors / Royal Blood	Combat: 94% Intel: 98% Agility: 72% Willpower: 90%	<i>'Let us aim for a perfect game.'</i>
Annie Leonhart Female Titan	Female Titan / Warrior Operative Marley Warriors / MP Infiltrator	Combat: 95% Intel: 86% Agility: 96% Willpower: 84%	<i>'I just... want to go back home.'</i>
Pieck Finger Cart Titan	Cart Titan / Strategic Scout Marley Warrior Unit	Combat: 80% Intel: 96% Agility: 92% Willpower: 88%	<i>'As expected of Pieck, exactly right.'</i>
Jean Kirstein Field Leader	Squad Leader / Field Commander Survey Corps Special Squad	Combat: 88% Intel: 89% Agility: 88% Willpower: 87%	<i>'There's something we have to do right now!'</i>

Sasha Braus Master Sniper	Master Sharpshooter / Vanguard Survey Corps Special Squad	Combat: 86% Intel: 70% Agility: 94% Willpower: 85%	'Meat...'
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5. WORLDBUILDING, TERRITORIAL WALLS & TACTICAL MAP

5.1 The Three Concentric Walls of Paradis

Constructed from millions of crystallized Colossal Titans by Karl Fritz in 743, the three walls enclose the remnant Eldian population on Paradis Island:

Wall Name	Radius & Height	Defensive Bastions	Tactical Role & Status
Wall Maria Outer Ring	Radius: 480 km Height: 50 meters	Shiganshina (South) Quinta (West) Holst (North) Rapier (East)	Outer defensive perimeter protecting 723,800 km² agricultural territory. Breached in 845; fully reclaimed in 850 by Survey Corps.
Wall Rose Middle Ring	Radius: 380 km Height: 50 meters	Trost (South) Krolva (West) Utopia (North) Karanes (East)	Middle line of defense containing residential and secondary industrial hubs. Gate attacked in 850; sealed with colossal boulder by Eren.
Wall Sina Inner Sanctuary	Radius: 250 km Height: 50 meters	Stohess (East) Hermiha (South) Yarckel (West) Orvud (North)	Innermost sanctuary protecting Mitras royal capital and high-ranking nobility. Heavily fortified by Military Police Brigade.

5.2 Vertical Maneuvering Equipment (ODM Gear) Specifications

The pinnacle of human anti-Titan mobility technology, powered by Iceburst Stone compressed gas:

- **Compressed Gas Turbines:** Back-mounted high-pressure canisters expelling volatile gas refined from volcanic caverns.
- **Dual Wire Launcher Drums:** Hip-mounted steel cable spools firing high-tensile anchor hooks (4,500 kg/cm²).
- **Ultra-Hard Steel Blades:** Flexible segmented carbon-steel alloy strips engineered specifically to slice through tough Titan nape flesh.
- **Operational Ceiling:** Enables soldiers to achieve velocities in excess of 80 km/h in forest and urban environments.

5.3 Tactical Satellite Map Locations

- **Shiganshina District (Coords 34.5° N, 12.8° E):** Site of the 845 breach, 850 reclamation battle, and Yeagerist final stand.
- **Trost District (Coords 32.1° N, 10.4° E):** Site of humanity's first victory over Titans using Eren's boulder sealing maneuver.
- **Stohess District (Coords 31.8° N, 14.2° E):** Urban battleground where the Female Titan was unmasked within Wall Sina.
- **Giant Forest (Coords 33.4° N, 11.7° E):** 80-meter trees providing ideal anchor terrain for Survey Corps ambushes.
- **Fort Salta (Coords 18.2° S, 45.1° W):** Final cliffside bastion in Marley where the Rumbling was halted in the Battle of Heaven and Earth.

6. COMPLETE CODEBASE EVOLUTION & GIT COMMIT CHRONOLOGY

The following log details the step-by-step transformation of the codebase from scratch up to commit **b604d57**:

Commit Hash	Commit Message	Architectural Enhancement
Initial	Create full 5-page AOT interactive web application	Created index, timeline, titans, characters, and world portals with base datasets.
8b9fbd8	Remove audio soundboard and sound toggle	Purged audio controllers to ensure lightweight, zero-latency asset loading.
b234282	Add authentic Japanese quotes and romanizations	Added original Japanese voice lines and romanji to characters and timeline events.
5f379c3	Restore Wings of Freedom SVG military shield favicon	Vector SVG favicon created with overlapping blue/white feather plumes and gold border.
1c101e2	Update themes to True Pitch-Black OLED editions	Upgraded theme engine to support OLED Stealth, Titan Emerald, and Blood Black.
a9b590c	Set Blood Black as the permanent default theme	Removed theme toggle; locked app to true #000000 black, neon red, and smoked ruby glass.
c79fee5	Add enhanced Scroll Trigger system	Added Back-to-Top trigger, staggered reveals, and animated stat counters.
58b6b61	Make scroll trigger reveals dramatically evident	Removed top progress bar; implemented high-impact 80px glide and spring cubic curve.
b604d57	Remove blur effect from scroll triggers	Eliminated filter: <code>blur(...)</code> to maintain 100% razor-sharp card and typography clarity.

OFFICIAL VERIFICATION & ARCHIVE CERTIFICATE

This document certifies that the **Attack on Titan Lore Compendium & Tactical Archive** at Commit b604d57 contains the full canonical dataset for the 2,000-Year Timeline, the Nine Titan Shifters, 12 Military Personnel Dossiers, the Three Concentric Walls Radar, the ODM Gear technical breakdown, and the high-impact razor-sharp scroll trigger animation system.

Repository: <https://github.com/barnwaladitya66-eng/Attack-On-Titan.git>
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